Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/13/2021
PI Name:	Bellan, Paul Ph.D.		
Project Title:	Development of Laser Induced Fluorescence as a Diagnostic for Measuring Neutral, Ion, and Molecular Particle Fluxes in the PK-4 Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91125-0001	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	05/01/2017	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project has closed per PI; final reporting submitted (Ed., 5/2/22) NOTE: Project reinstated and extended to 12/31/2023 per U. Israelsson/JPL (Ed., 1/12/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1573432		
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Task Description:	The International Space Station (ISS) has onboard the PK-4 dusty plasma experiment which has the goal of studying dusty plasma behavior in zero-gravity conditions. Conventional plasmas are a gas composed of interacting electrons (small and negatively charged) and ions (large and positively charged) and, in addition, either small or large quantities of neutral particles. The electrons and ions interact with each other via electrical forces and exhibit complex behavior. A dusty plasma has electrons, ions, neutrals, and in addition, macroscopic charged particles ("dust") having dimensions of a few microns so there are now three types of charged particles. The dust grains are large enough to be visible and they organize themselves into complex patterns as a result of a balance of forces from the electrons, the ions, the neutral particles, and the other dust grains. The task of this program is to measure the ion and neutral flow velocities so as to determine the forces on the dust grains from these flows. The method to be used is laser induced fluorescence. This method in principle provides a direct measurement of the neutral or ion flow velocity. It works by illuminating the neutrals or ions with a laser tuned to a carefully chosen wavelength so that bound electrons on the neutrals or ions are excited to a higher atomic energy state. The excited neutrals or ions quickly decay to a lower energy state and emit a photon that is at a wavelength different from that of the exciting laser. The emitted (fluorescent) photon is picked up by a sensitive photodetector. The wavelength of the exciting laser needs to be extremely precise in order for the scheme to work and most importantly, if the neutral or ion being illuminated is moving, the laser wavelength appears shifted in the frame of this moving neutral or ion. The tuning of the exciting laser then has to be changed to compensate for this wavelength shift. By noting the amount of wavelength compensation that is required, the velocity of the moving neutral or ion can be determined. The dusty plasma experiment at Caltech is being used to develop this method for measuring ion and neutral velocities in a dusty plasma.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Until now, the neutral and ion flow velocity in dusty plasma has been estimated using theoretical analysis but there have been no direct measurements. The measurements to be made will indicate whether the forces on the dust grains in a dusty plasma are indeed what has been presumed. The measurement is remote and non-perturbing and should be able to provide flow patterns. If successful, the measurements will provide useful information on the ion and neutral flow patterns in the PK-4 dusty plasma experiment on board the International Space Station (ISS). Dusty plasmas exist in several space contexts such as the diffuse rings of Saturn, comet tails, and accretion disks orbiting new-born stars. The agglomeration of accretion disks is the beginning stage of planet formation and so it is likely that at some point in time the material constituting Earth was orbiting the Sun as a dusty plasma. Determining the mechanisms governing dusty plasma behavior will provide improved understanding of these contexts. The dust in the Caltech experiment consists of small grains of water ice and so may provide direct link to the space dusty plasmas which in many cases (e.g., Saturn rings, comet tails) are also composed of water ice.
Task Progress:	SUPPLEMENTAL REPORTING - NOVEMBER 2022 As a further follow-on, I was provided with funding sufficient to provide the majority of support for two Caltech undergraduates to be Summer Undergraduate Research Fellows (SURF) during the summer of 2022. Undergraduate Joshua Pawlak did a SURF developing a terahertz detector based on a field effect transistor. The project was ultimately successful and culminated in detection of a 500 GHZ test signal at a JPL lab doing terahertz technology. Undergraduate Geoff Pomraning constructed and tested a special plasma density diagnostic for use in the dusty plasma experiment in my lab. This diagnostic involved generating an electron beam from a biased, heated, emissive filament probe and then detecting oscillations at the plasma frequency that were generated by a beam-plasma instability. The frequency of the oscillations were determined with an rf spectrum analyzer and gave a direct measurement of the plasma density as the frequency of the oscillations depends only on the plasma density. Both Joshua Pawlak and Geoff Pomraning presented the results of their SURF projects in person at the annual American Physical Society Division of Plasma Physics meeting which was held in Spokane during October 17-21, 2022. Geoff Pomraning was recognized with an Outstanding APS Undergraduate Poster Award. Joshua Pawlak is being nominated for a Perpall speaking award for his talk at the SURF seminar day and so will participate in the Perpall speaking award competition to be held at Caltech in January 2023. SUPPLEMENTAL REPORTING - AUGUST 2021 Purpose 1. To develop laser induced fluorescence (LIF) as a diagnostic for dusty plasmas 2. To use LIF to measure neutral and ion flows 3. To determine how neutral and ion flows affect dusty plasma and are affected by the dust. Accomplishments Major Activities: Develop laser induced fluorescence diagnostic for use in a dusty plasma and measure behavior of special interest to dusty plasma physics. Specific Objectives: Measure neutral and ion flow velocities in a dusty plasma using laser induced fluorescence. Investigate coupling of flow to dust behavior. Significant Results: A method for measuring neutral flow was successfully developed. The neutral flow measurement involved measuring the vector velocity in two dimensions over a large grid so the pattern of flow from a nozzle was well resolved. The method was extremely challenging because the flow velocity was much slower than expected. Doppler wavelength shifts of femtometers were resolved to achieve these measurements. Key Outcomes or Other Achievements:

Doppler shifts of laser lines corresponding to velocities less than one meter per second were resolved (shifts of about three parts in a billion).

The dust grains were pushed by the flow but would revert back to their original position when the flow was turned off.

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Relevant equipment was ordered, acquired, and installed. This includes a Toptica laser controller, a laser head for an argon ion line and a laser head for an argon neutral line.

Attempts to observe the argon ion fluorescence line were not successful which was unexpected since argon lines had been previously observed. It was determined that the much higher neutral pressure in this experiment was quenching the ion metastables necessary for fluorescence.

In contrast, attempts to observe the argon neutral fluoresence were very successful and a strong signal was obtained. This signal served for the basis of developing a sophisticated system that measured the neutral flow velocity. The measurement required resolving the extremely small Doppler shifts resulting from neutrals moving of the order of 1 meter per second. The motion of the neutrals required the laser to be detuned from the resonant frequency by about 3 parts in a billion (1 m/s compared to speed of light). This Doppler shift was about an order of magnitude less than the typical laser drift and so a double subtraction system was used to remove the laser drift. The laser beam was split into two beams which were arranged to traverse the plasma in opposite directions and so would have opposite Doppler shifts. A chopper and lock-in amplifier was used to measure the fluorescent signal and the laser frequency was scanned to map out the neutral velocity distribution profile. The center of this profile was slightly shifted as a result of the flow and this fit was determined by fitting a Gaussian distribution (bell curve). The difference between the Gaussian center for the two beam directions gave the Doppler shift with the laser drift subtracted out.

By having two gas nozzles at 45 degree angles and taking separate measurements for each nozzle for a two-dimensional array of positions, the velocity components parallel and perpendicular to the laser beam were measured. These were then used to produce a two-dimensional map of the velocity vectors. This map provided a clear indication of the flow from a nozzle and showed the flow slowing down and curving with increasing distance from the nozzle. The velocity was shown to be slower when there was a higher background pressure and the interaction of the flow with ice grain dusty plasma was observed. It was seen that a group of ice grains was pushed away by the flow but would then revert to the original location when the flow was turned off.

Bibliography Type:	Description: (Last Updated: 03/29/2021)
Articles in Peer-reviewed Journals	Marshall RS, Bellan PM. "Laser-induced fluorescence measurement of very slow neutral flows in a dusty plasma experiment." Rev Sci Instrum. 2020 Jun 1;91(6):063504. https://doi.org/10.1063/5.0006684 ; PMID: 32611057 , Jun-2020
Awards	Marshall RS, Bellan PM. "Laser-induced fluorescence measurement of very slow neutral flows in a dusty plasma experiment. Rev Sci Instrum. 2020 Jun 1;91(6):063504. This paper was selected to be an 'Editor's Pick' by the journal." Jun-2020